Submodule 35.01: What is Git & GitHub?

1. Introduction to Version Control

In software engineering and data science, projects are rarely built in a single sitting or by a single individual. Real-world machine learning systems, web platforms, and production pipelines evolve continuously across weeks, months, or years.

Before modern version control, developers tracked code versions manually:

```
my_ml_project_v1.py
my_ml_project_v2_final.py
my_ml_project_v2_final_FINAL.py
my_ml_project_v3_working_dont_touch.py
```

This manual approach leads to catastrophic issues:

- **Zero Visibility**: No record of what changes were made, why they were made, or who made them.
- **Overwritten Work**: Multiple developers editing the same file concurrently overwrite each other's code.
- **No Safe Rollback**: If a newly added feature crashes the production system, restoring previous working state is nearly impossible.

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2. What is Git?

Git is an open-source, distributed **Version Control System (VCS)** designed to track changes in source code across time. It was created in 2005 by **Linus Torvalds** (the creator of the Linux operating system kernel).

Core Capabilities of Git:

1. **Change Tracking**: Every modification to any file in your project is tracked with surgical precision.
2. **Timeline Snapshots (Commits)**: Developers create explicit checkpoints (commits) capturing the exact state of the entire project.
3. **Time Travel & Rollback**: You can inspect or roll back your project to any historical commit at any time.

4. **Branching & Parallel Workflows**: Multiple team members can work on isolated feature branches simultaneously without disrupting the main production code.
5. **Distributed Architecture**: Every developer possesses a complete local copy of the repository and its entire history on their computer, allowing offline work.

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3. What is GitHub?

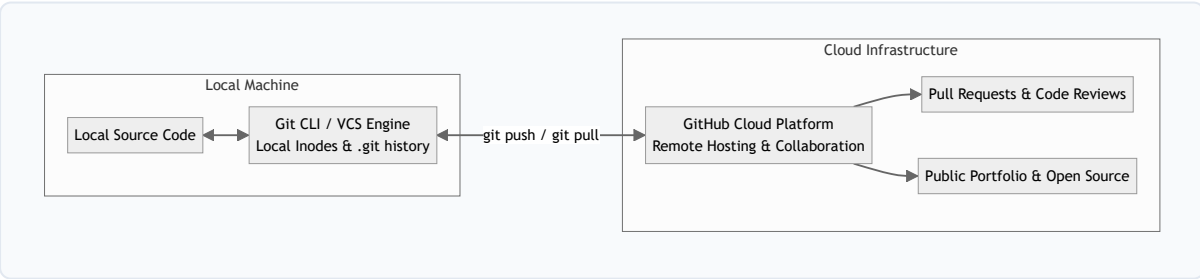
GitHub is a cloud-based hosting platform and service built on top of Git. Owned by Microsoft, GitHub allows developers to store, manage, share, and collaborate on Git repositories over the internet.

Core Capabilities of GitHub:

1. **Cloud Repository Hosting**: Serves as a centralized remote backup and synchronization hub for local Git repositories.
2. **Collaboration Features**: Facilitates team coordination via **Pull Requests (PRs)**, Code Reviews, Issue Tracking, and Discussions.
3. **Developer Portfolio**: A developer's GitHub profile showcases their public repositories, code contributions, open-source work, and commit activity graph to tech recruiters and engineering managers.
4. **CI/CD & Automation**: Integrated workflows (GitHub Actions) automate testing, building, and deployment of machine learning models and applications.

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4. Key Differences: Git vs GitHub



Dimension	Git	GitHub
Fundamental Nature	Version Control System software	Cloud hosting platform & web service
Installation	Installed locally on your computer	Accessed via web browser (`github.com`)

Dimension	Git	GitHub
Creator / Owner	Linus Torvalds (Open Source, 2005)	Microsoft (Acquired in 2018)
Internet Requirement	Fully functional offline	Requires active internet connection
Interface	Command Line Interface (CLI) / Shell	Graphical Web Interface & REST API
Primary Responsibility	Tracking file changes, commits, branches	Remote backup, team collaboration, PR reviews